November 4, 2026

Editor-in-Chief
Spatial Statistics

Dear Editor,

Please consider the manuscript "Paired spatial statistics under alternative embeddings: exact graph-count variance and a prime-pattern application" for publication in *Spatial Statistics*.

The paper addresses comparison of spatial summaries when the same selected labels are mapped to different locations. A paired random-labelling procedure is described, and an exact finite-population variance is derived for a contrast of normalised edge counts on two simple graphs. The expression incorporates graph overlap and differences in vertex degrees, covering irregular open-window graphs as well as a simpler regular-toroidal case. The derivation is checked by exhaustive small-graph enumeration and paired simulations. A power-calibration study under planted gap-class enrichment, with fixed Gaussian thresholds and an independent second-seed level check, confirms that the paired procedure retains its nominal level and attributes the planted structure to the embedding that carries it.

The method is illustrated using three embeddings of the same finite prime set and is additionally applied to an open seismic dataset: earthquake epicentres rasterised onto cells over California are compared under rook and queen adjacency on the same cell population, with the exact null variance and a paired Monte Carlo test. Nearest-neighbour fits and established spatial summaries are compared with fixed-count and mapped inhomogeneous nulls. The manuscript also reports the substantial discrepancies between the continuous Brody fit and the tied lattice-distance distributions, and limits its conclusions to the specified models and finite windows.

The source code, numerical results and instructions needed to reproduce the analysis and figures are supplied with the manuscript. AI-assisted code and manuscript preparation are disclosed in the manuscript.

Thank you for considering the work.

Sincerely,

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